

1.5 Classwork Solutions: Applications

Notation: $=$ joins values that are exactly equal; $\approx$ introduces a rounded value. Each conversion runs as one chain of unit factors and is evaluated once, so there is nothing to round in the middle.

1. Rain on Manhattan.

$$(a) V = 20,000 \text{ m} \times 3000 \text{ m} \times 0.025 \text{ m} = \mathbf{1.5 \times 10^6 \text{ m}^3}$$

Converting each length to meters first avoids the square-kilometer trap: 1 km^2 is 10^6 m^2 , not 1000 m^2 .

$$(b) 1.5 \times 10^6 \text{ m}^3 \times \frac{1000 \text{ L}}{1 \text{ m}^3} \times \frac{1 \text{ kg}}{1 \text{ L}} = \mathbf{1.5 \times 10^9 \text{ kg}}$$

$$1.5 \times 10^9 \text{ kg} \times \frac{1 \text{ t}}{1000 \text{ kg}} = \mathbf{1.5 \times 10^6 \text{ metric tons}}$$

$$(c) 1.5 \times 10^6 \text{ m}^3 \times \frac{1000 \text{ L}}{1 \text{ m}^3} \times \frac{1 \text{ gal}}{3.785 \text{ L}} = 3.963 \dots \times 10^8 \text{ gal} \approx \mathbf{4.0 \times 10^8 \text{ gal}}$$

About **0.4 of a day's supply**, or roughly 10 hours of the whole city's water use — from one storm, on one borough.

Two figures throughout, from the 2.5 cm and 3.0 km (the 20 km is a rough rectangle too). Start the gallons from the volume in part (a), not from a rounded intermediate.

2. The oil slick.

$$(a) 1.0 \text{ L} \times \frac{1 \text{ m}^3}{1000 \text{ L}} = \mathbf{1.0 \times 10^{-3} \text{ m}^3}$$

$$(b) A = \frac{V}{\text{thickness}} = \frac{1.0 \times 10^{-3} \text{ m}^3}{2 \times 10^{-10} \text{ m}} = \mathbf{5 \times 10^6 \text{ m}^2}$$

Dividing the powers of ten: $10^{-3-(-10)} = 10^7$, and $1.0/2 = 0.5$. The units work too: $\text{m}^3/\text{m} = \text{m}^2$.

$$(c) \pi r^2 = 5 \times 10^6 \text{ m}^2, \text{ so } r = \sqrt{\frac{5 \times 10^6 \text{ m}^2}{\pi}} = 1261.5 \dots \text{ m and } d = 2r = 2523 \dots \text{ m} \approx \mathbf{2.5 \text{ km}} \text{ (about 3 km)}$$

The molecule size has one significant figure, so this is an estimate: “a couple of kilometers” is the physics. Franklin's teaspoon covered about half an acre. In 1890 Lord Rayleigh repeated the experiment carefully and used it to make one of the first estimates of the size of a molecule.

3. Walking around the Earth.

$$(a) 40,075 \text{ km} \times \frac{1 \text{ h}}{5.0 \text{ km}} \times \frac{1 \text{ day}}{12 \text{ h}} = 667.91 \dots \text{ days} \approx \mathbf{670 \text{ days}}$$
, better written $\mathbf{6.7 \times 10^2 \text{ days}}$

Two figures, set by the 5.0 km/h. The 12 hours is a stated assumption, not a measurement.

$$(b) \ 667.91 \text{ days} \times \frac{1 \text{ yr}}{365 \text{ days}} = 1.829 \dots \text{ yr} \approx \mathbf{1.8 \text{ years}}$$

Carry the unrounded 667.91 into this step. And the Pacific Ocean is in the way.

4. Sunlight.

$$t = 1.496 \times 10^8 \text{ km} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ s}}{3.00 \times 10^8 \text{ m}} \times \frac{1 \text{ min}}{60 \text{ s}} = 8.3111 \dots \text{ min} \approx \mathbf{8.31 \text{ min}}$$

Three figures, set by the 3.00. Rearranging $v = d/t$ to $t = d/v$ is the same thing as the $\frac{1 \text{ s}}{3.00 \times 10^8 \text{ m}}$ factor — the speed of light is a conversion factor between meters and seconds. The Sun we see is the Sun of eight minutes ago.